MOTOR GRADERS

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APPLICATIONS, Motor Graders:

The broad line of Cat motor graders allows the customer to choose a motor grader that best fits the intended application. Below is a summary of the typical motor grader applications.

Finish Grading

This application involves preparing a roadway or site surface for future paving or other construction activity. The material being moved is usually a hard, dry base material on a solid underfoot. Finish blading is the motor grader application that requires the highest degree of accuracy. Thus, it is primarily done at low operating speeds — usually less than 5 km/h (3 mph) — in gears 1 and 2. To ensure a smooth, even finished surface, one gear is usually maintained for a given pass. Pass lengths during this application are usually less than 600 m (2000 feet) for road construction and 150 m (500 feet) for site development. Most finish blading is performed by contractors in the Heavy Construction and Building Construction industries.

Heavy Blading

This application involves cutting, moving, and mixing material, usually in the initial stages of surface preparation. A variety of material types are moved in this manner, and the blade tip position varies accordingly. Full blade loads are usually experienced during heavy blading, since moving material is the primary goal. Pass lengths within this application vary, but are usually less than 600 m (2000 feet). Unlike finish blading, the speed of the machine is dependent on the load being moved when heavy blading material. Typical operating speeds are from 0-10 km/h (0 to 6 mph). Therefore, gears 2 through 4 are frequently used in this application. Most heavy blading activity is performed by contractors in the Heavy Construction, Governmental, Industrial, and Forestry industries.

Site Preparation

This application involves any material cutting, moving, and mixing necessary to prepare a residential, commercial, or industrial site for construction. A variety of materials are encountered in this application. Blade loads vary depending on the activity being performed. Both heavy blading and finish blading are performed when preparing a site. Pass lengths are typically in the range of 30-300 m (100 to 1000 feet). Typical operating speeds for site preparation vary depending on whether heavy blading or finish blading activities are being performed. Most site preparation activities are performed by contractors in the Building Construction industry.

Road Maintenance

This application involves reshaping dirt or gravel roads to maintain a crown or superelevation, or restoring the surface itself. This generally involves secondary roads maintained by governmental bodies such as townships and counties. Materials being moved in this application vary from extremely hard dirt bases to moist gravel surfaces. The typical blade load falls between that of finish blading and heavy blading. Pass lengths are frequently longer than 600 m (2000 feet) and can extend for miles. The general speed range for this application is 5-16 km/h (3 to 10 mph), corresponding to gears 2 (heavy dirt) through 5 (soft gravel). As with finish blading, accuracy of the graded surface is the primary concern in this application. Thus, frequent shifts should be avoided whenever possible. A gear should be chosen and maintained unless there is a significant change in the material being moved. Most road maintenance activities are performed by the Governmental industry.

Haul Road Maintenance

This application of the motor grader involves reshaping haul roads at mining, construction, or forestry work sites, usually for the purpose of maintaining smooth travel surfaces for equipment. Materials being moved while maintaining haul roads vary widely. Typical blade loads are about one-third to half of full capacity. Haul roads that experience large hauling units travelling on soft material may require heavy blade loads in order to reshape the road surface. Pass lengths vary depending on the application but can extend for miles on remote forestry or large mine haul roads. The general speed range for haul road maintenance is heavily dependent on the material being moved as well as the grade of the haul road. Many mine sites are in mountainous areas, requiring haul roads with steep grades. Generally, haul-road maintenance is performed at speeds similar to those required for general road maintenance 5-16 km/h (3 to 10 mph).

A travel surface that allows for the safe and efficient movement of machinery is the ultimate goal with this motor grader application. Very precise roadway elevations and slopes are desired but less crucial than when finish blading. Most haul road maintenance activities are performed by the Mining, Heavy Construction, and Forestry industries.

Side/Bank Slope Work

This application involves preparing side slopes or bank slopes along roadways by placing the moldboard on a sloped surface. Slopes of up to a 2:1 angle can be cut using a motor grader. Often the motor grader is operated on the level surface adjacent to the slope, and the moldboard is extended outward to the sloped surface. Fine soils are generally encountered in this application of the motor grader. Blade loads are usually less than half of the full blade capacity, and pass lengths are seldom longer than 600 m (2000 feet). A smooth-graded sloped surface is the primary concern in this application so frequent shifts should be avoided. The typical speed range is 0-6 km/h (0 to 4 mph), corresponding to a gear selection of 1 to 3. The nominal speed is heavily dependent on the type of material being moved and on the slope of the surface. Most side/bank slope work is performed by the Heavy Construction and Governmental industries.

Ditch Building/Cleaning

This application involves cutting "V" and flat-bottom ditches for drainage purposes and rebuilding them when necessary. Due to excessive rain and/or poor material, ditches often need cleaning and reshaping. When building ditches, materials with a wide range of densities are encountered. Blade loads vary accordingly, from half to full-blade capacity. Pass lengths are usually less than 600 m (2000 feet). The primary objective is to move material in a manner that yields a ditch with the desired slope. Ditch building often involves cutting and moving material of high density. Therefore, typical speed ranges vary. Most ditch building work, however, is performed in gears 1 through 3, corresponding to a maximum speed of about 8 km/h (5 mph). Ditch cleaning usually involves blading moist materials underneath a sod cover. Blade loads are usually less than half of full blade capacity when cleaning ditches, and pass lengths are similar to those encountered in ditch building. Typical maximum speeds for this activity are similar to that of ditch building, but less of a blade load is experienced. Ditch building and cleaning activities are usually performed by the Heavy Construction and Governmental industries.

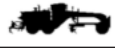
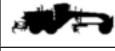
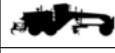
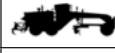
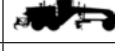
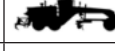
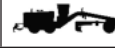
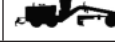
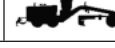
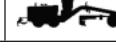
Ripping/Scarifying

This application involves conditioning hard, rough soils before they are bladed. Shanks on the ripper and/or scarifier are pushed into the ground, thus breaking up otherwise hard surfaces. Hard materials such as asphalt can also be loosened in order to make grading operations less damaging to the moldboard. Rippers and scarifiers can also be used to mix aggregates together. The materials being ripped/ scarified are usually hard and dry. Rippers generally penetrate 150-300 mm (6 to 12 inches) into the ground, while scarifiers typically penetrate to a depth of 25-200 mm (1 to 8 inches). Pass lengths are generally less than 600 m (2000 feet) for both activities. Since the material being ripped/scarified is generally hard, the typical maximum speed for this application is about 6 km/h (4 mph) gears 1-2. If the ripper/scarifier is used for mixing aggregates, the typical operating range becomes 6-20 km/h (4 to 12 mph) gears 3-6. Most ripping/scarifying activities are performed by the Heavy Construction and Governmental industries.

Snow Removal

Snow removal is the process of cutting and removing snow or ice from the roadway. In addition to the standard motor grader moldboard, other attachments such as a snow wing, V-plow, one-way plow, or reversible plow can be used to remove the snow. The moldboard itself is the most commonly used attachment for snow plowing. It is used in areas where snow depths are low, the terrain is relatively flat, and where excessive drifting does not occur. A snow wing is a moldboard that attaches to the machine's right side. The wing's curvature lifts the snow and "wings" it off the plowed surface. The snow wing is often used in conjunction with the standard moldboard, where the moldboard cuts the material and feeds it onto the wing. V-plows are mounted in front of the motor grader and are designed to dig into and lift packed snow. The typical speed range for snow removal is 10-30 km/h (6 to 18 mph), corresponding to a gear range of 3 to 7. Snow plowing often involves lower speeds than snow removal. The typical operating range for snow plowing is 8-19 km/h (5 to 12 mph) gears 2 to 4. The majority of Snow Removal/Plowing operations are performed by the Governmental, Mining, and Forestry industries.

TRUCK TO MOTOR GRADER MATCH

	740	770	775	777	785	789	793	797
12/140/160								
14								
16								
18								
24								

NOTE: Calculations based on 30 degree blade angle, standard moldboard width.
May not be applicable in all applications depending on haul road damage.
Rule of thumb 2.5 times the truck width.

PRODUCTION

The motor grader is used in a variety of applications in a variety of industries. Therefore, there are many ways to measure its operating capacity, or production. One method expresses a motor grader's production in relation to the area covered by the moldboard.

Formula:

$$A = S \times (L_e - L_o) \times 1000 \times E \quad \text{(Metric)}$$
$$A = S \times (L_e - L_o) \times 5280 \times E \quad \text{(English)}$$

where A: Hourly operating area (m^2/h or ft^2/h)
S: Operating speed (km/h or mph)
L_e: Effective blade length (m or ft)
L_o: Width of overlap (m or ft)
E: Job efficiency

Operating Speeds:

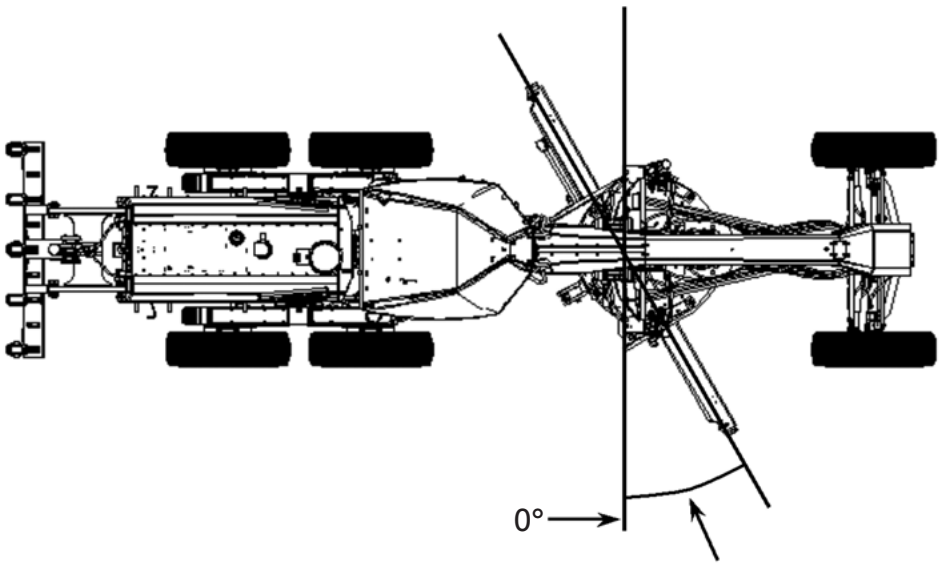
Typical operating speeds by application

Finish Grading:	0-4 km/h	(0-2.5 mph)
Heavy Blading:	0-9 km/h	(0-6 mph)
Ditch Repair:	0-5 km/h	(0-3 mph)
Ripping:	0-5 km/h	(0-3 mph)
Road Maintenance:	5-16 km/h	(3-9.5 mph)
Haul Road Maintenance:	5-16 km/h	(3-9.5 mph)
Snow Plowing:	7-21 km/h	(4-13 mph)
Snow Winging:	15-28 km/h	(9-17 mph)

Effective Blade Length:

Since the moldboard is usually angled when moving material, an effective blade length must be computed to account for this angle. This is the actual width of material swept by the moldboard.

NOTE: Angles are measured as shown below. The effective length becomes shorter as the angle increases.



Moldboard Angle

Moldboard Length, m (ft)	Effective Length, m (ft) 30 degree blade angle	Effective Length, m (ft) 45 degree blade angle
3.658 **(12)**	3.17 **(10.4)**	2.59 **(8.5)**
4.267 **(14)**	3.70 **(12.1)**	3.02 **(9.9)**
4.877 **(16)**	4.22 **(13.9)**	3.45 **(11.3)**
7.315 **(24)**	6.33 **(20.8)**	5.17 **(17.0)**

For other blade lengths and carry angles:
Effective length = COS [Radians (Blade L)] 3 Blade Length

Width of Overlap:

The width of overlap is generally 0.6 m (2.0 ft). This overlap accounts for the need to keep the tires out of the windrow on the return pass.

Job Efficiency:

Job efficiencies vary based on job conditions, operator skill, etc.

A good estimation for efficiency is approximately 0.70 to 0.85, but actual operating conditions should be used to determine the best value.

Example problem:

A Cat motor grader with a 3.66 m (12 ft) moldboard is performing road maintenance on a township road. The machine is working at an average speed of 13 km/h (8 mph) with a moldboard carry angle of 30 degrees. What is the motor grader's production based on coverage area?

Note: Due to the long passes involved in road maintenance — fewer turnarounds — a higher job efficiency of 0.90 is chosen.

Solution:

From the table, the effective blade length is 3.17 m (10.4 ft).

Metric

Production, A = 13 km/h × (3.17 m − 0.6 m) × 1000 × 0.90
= **30 069 m²/hr (3.07 hectares/hr)**

English

Production, A = 8 mph × (10.4 ft − 2.0 ft) × 5280 × 0.90
= **319,334 ft²/hr (7.33 acres/hr)**

To pinpoint the theoretical number of motor graders required to properly maintain your haul roads, based on your specific mining applications, please download the haul road maintenance calculator on *https://catminer.cat.com.*

Haul road maintenance impacts cycle time, tire, frame and drive train components, safety and ultimately your cost per ton. To achieve optimal truck productivity, your haul roads must be properly maintained.

Moderate:
- Road Maintenance
- Pad Cleaning
- Rock Clearing
- Shoulder Sweeping

Difficult:
- Ripping
- Spreading Dump Material
- Road Profiling/Reshaping

BLADE PULL

This specification is also known as drawbar pull. This spec can be calculated as follows:

Variables:

Rear weight of machine = Wr

Tire traction coefficient = T (Look up the table entitled "Coefficient of Traction Factors")

Wr × T = Blade Pull

Example problem:

Calculate the blade pull for a 140M Global Version version machine operating in a quarry pit...

Metric

RW = 10 501 kg

T = 0.65

10 501 × 0.65 = 6825.65

English

RW = 23,151 lb

T = 0.65

23,151 × 0.65 = 15,048.15

BLADE DOWN PRESSURE

This spec can be calculated as follows:

Variables:

Blade to front axle length = BA

Wheel base length = WB

Weight on front wheels = FW

Blade down pressure = BD

$$\frac{\mathbf{WB}}{\mathbf{(WB - BA)}} \times \mathbf{FW = BD}$$

Example problem:

Calculate the blade down pressure for a 140M Global Version version machine...

Metric

BA = 2565 mm FW = 4223 kg

WB = 6086 mm BD = ?

$$\frac{\mathbf{6086}}{\mathbf{(6086 - 2565)}} \times \mathbf{4223 = 7299\ kg}$$

English

BA = 101 in FW = 9310 lb

WB = 240 in BD = ?

$$\frac{\mathbf{240}}{\mathbf{(240 - 101)}} \times \mathbf{9310 = 16{,}075\ lb}$$

This specification is only a minor indicator of a motor grader's productivity. It alone gives no measure of overall machine productivity. When considering motor grader production you need an optimum balance between the machine's front and rear weights. If a machine has too much weight on the front axle, it might have a high blade down pressure spec. It will, however, lack the essential rear weight and traction needed to push through the load. Too much weight in the rear and it will not have the necessary weight in the front during heavy cuts to maintain proper steering control.

Cat machines are built with this optimum balance in mind. A Cat motor grader is engineered with the proper weight distribution necessary for maximum productivity.

Effective Blade Length*

		Moldboard							
		3.66 m (12')		4.27 m (14')		4.88 m (16')		7.32 m (24')	
		m	ft	m	ft	m	ft	m	ft
Angle°	0°	3.66	**12.00**	4.27	**14.00**	4.88	**16.00**	7.32	**24.00**
	5°	3.64	**11.95**	4.25	**13.95**	4.86	**15.94**	7.29	**23.91**
	10°	3.60	**11.82**	4.20	**13.79**	4.80	**15.76**	7.21	**23.64**
	15°	3.53	**11.59**	4.12	**13.52**	4.71	**15.45**	7.07	**23.18**
	20°	3.44	**11.28**	4.01	**13.16**	4.58	**15.04**	6.87	**22.55**
	25°	3.32	**10.88**	3.87	**12.69**	4.42	**14.50**	6.63	**21.75**
	30°	3.17	**10.39**	3.69	**12.12**	4.22	**13.86**	6.33	**20.78**
	35°	3.00	**9.83**	3.50	**11.47**	4.00	**13.11**	5.99	**19.66**
	40°	2.80	**9.19**	3.27	**10.72**	3.74	**12.26**	5.61	**18.39**
	45°	2.59	**8.49**	3.02	**9.90**	3.45	**11.31**	5.17	**16.97**

*Effective blade length is the amount of blade coverage the machine is capable of when the blade is at a given angle.

EXTREME SLOPE OPERATION

There are two ways of defining slope work. The slope perpendicular to the machine's direction of travel is commonly referred to as "Side Sloping." The slope parallel to the machine's direction of travel — the machines ability to travel up or down terrain, is commonly referred to as "Gradeability."

Side Sloping capability for our Cat graders is somewhat subjective, but general agreement among professional operators is that working on a slope ratio of 2.5:1 (21.8 degrees) is the safe limit … an experienced operator may be able to operate on a 2:1 (28 degrees) slope. Many factors influence this limit such as operator experience, machine configuration, tires and soil conditions, but a 2.5:1 is achievable. Further, a 3:1 slope is the approximate maximum side slope a grader can work on in straight frame configuration. The steeper side slopes all require the machine be articulated to safely navigate the slope.

Gradeability is approximately 22 degrees. This is established by the grader's ability to stop without skidding the tires while moving downhill. The motor grader can, however, *climb* grades steeper than 22 degrees. The traction coefficient is the critical factor in determining whether a grader can safely navigate the slope. Caterpillar recommends that you never climb a slope steeper than you can safely descend.

Maximum lubrication angle: We have measured the graders on a tilt table and pump cavitation occurs around 30 degrees (58% or 1.7:1). This is beyond the grade or slope a motor grader can operate on.

When working side hills and slopes, consideration should be given to the following important points.

- **Speed of Travel —** At higher speeds, inertia forces tend to make the grader less stable.
- **Roughness of Terrain or Surface —** Ample allowance should be made where the terrain or surface is uneven.
- **Mounted Equipment —** Mounted attachments such as front plows, snow wings, rippers and other mounted equipment cause the tractor to balance differently.
- **Nature of Surface —** New earthen fills may give way with the weight of the grader. Rocky surfaces may promote side slipping of grader.
- **Excessive Loads or Side Draft —** This may cause wheel slippage, where the downhill tires "dig in," increasing the angle of grader.
- **Tire Selection and Maintenance —** Consideration should be given to proper tire selection and air pressure. For more information, consult Caterpillar publications — Motor Grader Tire Selection Guide and Operation and Maintenance Manual.
- **Drawbar, Circle and Blade Position —** The position of the blade can affect the stability of the machine.
- **Articulation Angle —** Articulation angle can affect the stability of the machine.
- **Wheel Lean Angle —** Wheel lean angle can affect the stability of the machine.

NOTE: Safe operation on steep slopes may require special machine maintenance as well as excellent operator skill and proper equipment setup for the specific application. Consult Caterpillar publications for further operating tips — Operation & Maintenance Manual, Motor Grader Application Guide, and the Grade Comparison Chart in the Tables section of this Performance Handbook.

PAVING PRODUCTS

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Cold Planer

Sustainability:

- **Long-lasting lubricants and fluids** extend maintenance intervals, decreasing the amount of waste and filters contributed to environment.
- **Superior Cat® cutting tools last longer**, providing higher daily productivity with less fuel burn.
- **Available Cat Diamond Cutting Bits** last up to 80 times longer than conventional carbide bits.
- **Ecology drains** provide a simple means to drain machine fluids with a minimized risk of spillage.
- **Maintenance free Cat batteries** are recyclable.
- **Cat engines** meet applicable emission standards.
- **Cold planers recycle aggregate from worn out roads** for use in new roads, reducing the cost and energy needs required for excavating, processing and hauling virgin aggregate.

Features:

- **Cat engines.**
- **Up-cutting mandrels** provide cutting efficiency and improved bit life.
- **Excellent maneuverability** for productivity and job site flexibility.
- **Front-discharge conveyor** facilitates haul unit movement in congested urban applications.
- **Optimum weight-to-horsepower balance** for delivering maximum available horsepower to the cutter.
- **Computerized Monitoring System (CMS)** provides three warning levels for abnormal operating conditions.
- **Load control system** keeps machine operating at peak efficiency.
- **Water spray system** for dust control and bit cooling.
- **Optional Cat Grade and Slope** available for PM620 and PM622 models.

		Cutter/Drum Width — m²/min (yd²/min)									
Speed		**350 mm (1'2")**		**400 mm (1'4")**		**500 mm (1'8")**		**600 mm (2'0")**		**1000 mm (3'4")**	
m/min	**ft/min**	**m²**	**yd²**	**m²**	**yd²**	**m²**	**yd²**	**m²**	**yd²**	**m²**	**yd²**
3.0	**10**	1.1	**1.3**	1.2	**1.4**	1.5	**1.8**	1.8	**2.2**	3.0	**3.6**
4.6	**15**	1.6	**1.9**	1.8	**2.2**	2.3	**2.8**	2.8	**3.3**	4.6	**5.5**
6.1	**20**	2.1	**2.6**	2.4	**2.9**	3.1	**3.6**	3.7	**4.4**	6.1	**7.3**
7.6	**25**	2.7	**3.2**	3.0	**3.6**	3.8	**4.5**	4.6	**5.5**	7.6	**9.1**
9.1	**30**	3.2	**3.8**	3.6	**4.4**	4.6	**5.4**	5.5	**6.5**	9.1	**10.9**
10.7	**35**	3.7	**4.5**	4.3	**5.1**	5.4	**6.4**	6.4	**7.7**	10.7	**12.8**
12.2	**40**	4.3	**5.1**	4.9	**5.8**	6.1	**7.3**	7.3	**8.8**	12.2	**14.6**
13.7	**45**	4.8	**5.7**	5.5	**6.6**	6.9	**8.2**	8.2	**9.8**	13.7	**16.4**
15.2	**50**	5.3	**6.4**	6.1	**7.3**	7.6	**9.1**	9.1	**10.9**	15.2	**18.2**
16.8	**55**	5.9	**7.0**	6.7	**8.0**	8.4	**10.0**	10.1	**12.1**	16.8	**20.1**
18.3	**60**	6.4	**7.7**	7.3	**8.8**	9.2	**10.9**	11.0	**13.1**	18.3	**21.9**
Speed		**1220 mm (4'0")**		**1900 mm (6'3")**		**2010 mm (6'7")**		**2100 mm (6'11")**		**2210 mm (7'3")**	
m/min	**ft/min**	**m²**	**yd²**	**m²**	**yd²**	**m²**	**yd²**	**m²**	**yd²**	**m²**	**yd²**
3.0	**10**	3.7	**4.4**	5.7	**6.8**	6.0	**7.2**	6.3	**7.5**	6.6	**7.9**
4.6	**15**	5.6	**6.7**	8.7	**10.5**	9.2	**11.1**	9.7	**11.6**	10.2	**12.2**
6.1	**20**	7.4	**8.9**	11.6	**13.9**	12.3	**14.7**	12.8	**15.3**	13.5	**16.1**
7.6	**25**	9.3	**11.1**	14.4	**17.3**	15.3	**18.3**	16.0	**19.1**	16.8	**20.1**
9.1	**30**	11.1	**13.3**	17.3	**20.7**	18.3	**21.9**	19.1	**22.9**	20.1	**24.1**
10.7	**35**	13.1	**15.6**	20.3	**24.3**	21.5	**25.7**	22.5	**26.9**	23.6	**28.3**
12.2	**40**	14.9	**17.8**	23.2	**27.7**	24.5	**29.3**	25.6	**30.6**	27.0	**32.2**
13.7	**45**	16.7	**20.0**	26.0	**31.1**	27.5	**32.9**	28.8	**34.4**	30.3	**36.2**
15.2	**50**	18.5	**22.2**	28.9	**34.5**	30.6	**36.5**	31.9	**38.2**	33.6	**40.2**
16.8	**55**	20.5	**24.5**	31.9	**38.2**	33.8	**40.4**	35.3	**42.2**	37.1	**44.4**
18.3	**60**	22.3	**26.7**	34.8	**41.6**	36.8	**44.0**	38.4	**46.0**	40.4	**48.4**
Speed		**3050 mm (10'0")**		**3500 mm (11'6")**		**3810 mm (12'6")**					
m/min	**ft/min**	**m²**	**yd²**	**m²**	**yd²**	**m²**	**yd²**				
3.0	**10**	9.2	**10.9**	10.5	**12.6**	11.4	**13.7**				
4.6	**15**	14.0	**16.8**	16.1	**19.3**	17.5	**21.0**				
6.1	**20**	18.6	**22.3**	21.4	**25.5**	23.2	**27.8**				
7.6	**25**	23.2	**27.7**	26.6	**31.8**	29.0	**34.6**				
9.1	**30**	27.8	**33.2**	31.9	**38.1**	34.7	**41.5**				
10.7	**35**	32.6	**39.0**	37.5	**44.8**	40.8	**48.8**				
12.2	**40**	37.2	**44.5**	42.7	**51.1**	46.5	**55.6**				
13.7	**45**	41.8	**50.0**	48.0	**57.3**	52.2	**62.4**				
15.2	**50**	46.4	**55.4**	53.2	**63.6**	57.9	**69.3**				
16.8	**55**	51.2	**61.3**	58.8	**70.3**	64.0	**76.6**				
18.3	**60**	55.8	**66.8**	64.1	**76.6**	69.7	**83.4**				

NOTE: Above figures are based on approximately 25 mm **(1 in)** depth of cut. For greater depths of cut, multiply the production rate by cutting depth. Figures are based on asphalt density of 2322 kg/m³ **(145 lb/ft³)**. (As referenced on *AsphaltPavement.org* — the website of NAPA.)

Speed		Cutter/Drum Width — metric tons/min (U.S. tons/min)									
		350 mm (1'2")		400 mm (1'4")		500 mm (1'8")		600 mm (2'0")		1000 mm (3'4")	
m/min	ft/min	Metric tons	U.S. tons	Metric tons	U.S. tons	Metric tons	U.S. tons	Metric tons	U.S. tons	Metric tons	U.S. tons
3.0	**10**	0.06	**0.07**	0.07	**0.08**	0.09	**0.10**	0.10	**0.12**	0.17	**0.19**
4.6	**15**	0.09	**0.10**	0.11	**0.12**	0.13	**0.15**	0.16	**0.18**	0.27	**0.29**
6.1	**20**	0.12	**0.14**	0.14	**0.16**	0.18	**0.20**	0.21	**0.23**	0.35	**0.39**
7.6	**25**	0.15	**0.17**	0.18	**0.19**	0.22	**0.24**	0.26	**0.29**	0.44	**0.49**
9.1	**30**	0.18	**0.20**	0.21	**0.23**	0.26	**0.29**	0.32	**0.35**	0.53	**0.58**
10.7	**35**	0.22	**0.24**	0.25	**0.27**	0.31	**0.34**	0.37	**0.41**	0.62	**0.68**
12.2	**40**	0.25	**0.27**	0.28	**0.31**	0.35	**0.39**	0.42	**0.47**	0.71	**0.78**
13.7	**45**	0.28	**0.31**	0.32	**0.35**	0.40	**0.44**	0.48	**0.53**	0.80	**0.88**
15.2	**50**	0.31	**0.34**	0.35	**0.39**	0.44	**0.49**	0.53	**0.58**	0.88	**0.97**
16.8	**55**	0.34	**0.38**	0.39	**0.43**	0.49	**0.54**	0.59	**0.64**	0.98	**1.07**
18.3	**60**	0.37	**0.41**	0.42	**0.47**	0.53	**0.59**	0.64	**0.70**	1.06	**1.17**

Speed		1220 mm (4'0")		1900 mm (6'3")		2010 mm (6'7")		2100 mm (6'11")		2210 mm (7'3")	
m/min	ft/min	Metric tons	U.S. tons	Metric tons	U.S. tons	Metric tons	U.S. tons	Metric tons	U.S. tons	Metric tons	U.S. tons
3.0	**10**	0.21	**0.23**	0.33	**0.36**	0.35	**0.39**	0.37	**0.40**	0.38	**0.42**
4.6	**15**	0.33	**0.36**	0.51	**0.56**	0.54	**0.59**	0.56	**0.62**	0.59	**0.65**
6.1	**20**	0.43	**0.48**	0.67	**0.74**	0.71	**0.78**	0.74	**0.82**	0.78	**0.86**
7.6	**25**	0.54	**0.59**	0.84	**0.92**	0.89	**0.98**	0.93	**1.02**	0.98	**1.07**
9.1	**30**	0.64	**0.71**	1.00	**1.11**	1.06	**1.17**	1.11	**1.22**	1.17	**1.29**
10.7	**35**	0.76	**0.84**	1.18	**1.30**	1.25	**1.38**	1.30	**1.44**	1.37	**1.51**
12.2	**40**	0.86	**0.95**	1.35	**1.48**	1.42	**1.57**	1.49	**1.64**	1.57	**1.72**
13.7	**45**	0.97	**1.07**	1.51	**1.67**	1.60	**1.76**	1.67	**1.84**	1.76	**1.94**
15.2	**50**	1.08	**1.19**	1.68	**1.85**	1.77	**1.95**	1.85	**2.04**	1.95	**2.15**
16.8	**55**	1.19	**1.31**	1.85	**2.04**	1.96	**2.16**	2.05	**2.26**	2.16	**2.38**
18.3	**60**	1.30	**1.43**	2.02	**2.22**	2.14	**2.35**	2.23	**2.46**	2.35	**2.59**

Speed		3050 mm (10'0")		3500 mm (11'6")		3810 mm (12'6")	
m/min	ft/min	Metric tons	U.S. tons	Metric tons	U.S. tons	Metric tons	U.S. tons
3.0	**10**	0.53	**0.59**	0.61	**0.67**	0.66	**0.73**
4.6	**15**	0.81	**0.90**	0.93	**1.03**	1.02	**1.12**
6.1	**20**	1.08	**1.19**	1.24	**1.37**	1.35	**1.49**
7.6	**25**	1.35	**1.48**	1.54	**1.70**	1.68	**1.85**
9.1	**30**	1.61	**1.78**	1.85	**2.04**	2.01	**2.22**
10.7	**35**	1.89	**2.09**	2.17	**2.40**	2.37	**2.61**
12.2	**40**	2.16	**2.38**	2.48	**2.73**	2.70	**2.97**
13.7	**45**	2.43	**2.67**	2.78	**3.07**	3.03	**3.34**
15.2	**50**	2.69	**2.97**	3.09	**3.40**	3.36	**3.70**
16.8	**55**	2.97	**3.28**	3.41	**3.76**	3.72	**4.09**
18.3	**60**	3.24	**3.57**	3.72	**4.10**	4.05	**4.46**

NOTE: Above figures are based on approximately 25 mm **(1 in)** depth of cut. For greater depths of cut, multiply the production rate by cutting depth. Figures are based on asphalt density of 2322 kg/m^3 **(145 lb/ft^3)**. (As referenced on *AsphaltPavement.org* — the website of NAPA.)

MACHINE SELECTION

Prime considerations in selecting the proper cold planer model are:
— specifics of work to be done
— type of projects generally done by the contractor
● City/Urban or Highway/Airport
— desired production capacities

PM310/PM312/PM313 Cold Planers

The Cat PM310/PM312/PM313 Cold Planers feature compact dimensions and excellent maneuverability ideal for easy operation in urban applications. The machine performs controlled, full-depth removal of asphalt layers in a single pass and is also capable of concrete removal. The machine is available with a wheel or track undercarriage and features four steering modes for high maneuverability.

PM620 and PM622 Cold Planers

The Cat PM620 and PM622 Cold Planers are high-production, half-lane milling machines with excellent maneuverability and plenty of power to perform controlled full-depth removal of asphalt and concrete pavements in a single pass. The track-driven machine features four steering modes for high maneuverability. The PM620 features a 2.0 m (79") cutting width and performs well in urban environments or in applications where great maneuverability is required. The PM622 features a 2.2 m (88") cutting width and is ideal for high-production applications such as main line milling.

Cold Planer Characteristics (Highway/Airport)

Highway/Airport work requires high-volume cold planers. The PM620, PM622 and other high horsepower half-lane cold planers are being used more on Highway/Airport projects. Users like to have one machine that can work successfully on high production jobs then switch to city/urban applications.

Cold Planer Characteristics (City/Urban)

All Cat cold planers are four-track, front discharge models. Front discharge cold planers make traffic control easier in congested quarters. The trucks travel forward in the same direction as the cold planer. The trucks move in and out of traffic faster increasing production.

COLD PLANING FUNDAMENTALS

Definition

Cold planing is automatically controlled cold milling to restore the pavement surface to a specified grade and slope; remove bumps, ruts, and other imperfections; and leave a textured surface which can be opened immediately to traffic or overlayed with new pavement materials.

Production and Tooth Wear

Because pavement materials vary, so do production and tooth wear. While predicting the exact production rate and tooth wear on a particular job is difficult, general guidelines are available.

Production depends on the milling rate (the speed at which the cold planer moves forward). The machine's forward speed is determined, primarily, by aggregate type, asphalt bond strength and depth of cut. When milling asphalt pavement, the cold planer's teeth essentially are breaking the bond between asphalt-coated aggregate, not actually fracturing the aggregate itself. A pavement made with a mix containing a high percentage of fine aggregate and a high asphalt content is more difficult to mill than a pavement with a high percentage of coarse aggregate.

A dense or fine mix usually requires more power at the cutting drum, limiting the cold planer's forward speed. Decreased speed lowers production, and the tough bond between the small aggregate particles causes increased cutting-tooth wear. Lower production and higher tooth wear result in increased unit costs.

Cutting depth affects power demand at the drum and helps determine the cold planer's forward speed. However, production increases, to a point, as the depth of cut increases. For example, changing from a 25 mm (1 in) cut to a 51 mm (2 in) cut slows the machine only slightly but doubles the amount of material produced.

As the cut increases beyond the machine's peak-production depth, the reduced forward speed begins to offset the production gains of the deeper cut. For example, production at a 152 mm (6 in) cutting depth and slow speed may be no greater than cutting at a 76 mm (3 in) depth and a much faster speed.

Low Density Asphalt

Depth		PM620, PM622	
mm	**in**	**m/min**	**ft/min**
51	**2**	40	**130**
101	**4**	32	**105**
127	**5**	21	**70**
178	**7**	12	**40**
254	**10**	6	**20**
305	**12**	5	**15**

High Density Asphalt

Depth		PM620, PM622	
mm	**in**	**m/min**	**ft/min**
51	**2**	32	**105**
101	**4**	26	**85**
127	**5**	18	**60**
178	**7**	9	**30**
254	**10**	5	**15**
305	**12**	4	**12**

As long as the cold planer maintains a productive forward speed, deeper cuts will yield greater production and tend to lower tooth cost. Tooth wear does not increase in direct proportion to production when the machine is working in an efficient range.

Tooth wear at various depths for a given material is affected by how long the tooth remains in the cut. Because the teeth are mounted on a circular drum, each tooth cuts through the pavement in an arc. The tooth arc at a 102 mm (4 in) cutting depth, however, is not four times longer than at a 25 mm (1 in) cutting depth, even though production may be four times greater. The cutting arc at 102 mm (4 in) is approximately twice as long as that at 25 mm (1 in).

The peak cutting depth for a particular cold planer on a specific job is best determined by examining production, and subsequent costs, of a single deep cut versus multiple passes at a shallow depth.

APPLICATIONS

Although new applications for cold planers are being discovered, most work can be classified in six general categories:

Leveling and Bonding

This application removes a layer of pavement to eliminate potholes, ruts, bumps and other surface imperfections. The cold planer leaves a level, textured surface ideal for bonding to a new, thin overlay of asphalt or concrete. The surface has an interlocking texture with double the bonding area of a conventional smooth pavement. The textured surface and overlay form a monolithic bond, eliminating the shear plane that causes pavement layers to move and separate. Thinner overlays can be used, making the technique more economical than traditional overlay methods.

Surface Refinishing

Rough pavement can also be cold planed to specified grade and slope, providing a new riding surface without adding new paving materials. This application is particularly useful when base and sub-base are in good shape, or when several layers have been added to the roadway over the years. Roads can be cold planed during cold, wet months and reopened immediately. New overlays can be added whenever weather permits. This lengthens the practical working season for many contractors. The cold planer can also be used to correct expansion joint faults and pavement cracks.

Surface Repair

This category generally requires deeper cutting than leveling. It consists of removing isolated distressed pavement sections down to subbase, if necessary, prior to adding new overlay materials. Since the cutter mandrel on Cat cold planers cuts forward and upward, there's no damaging impact to the underlying base.

Pavement Removal

Pavement buildup is a problem that plagues most older streets, roads and highways. As overlays are added, curbs and drains are buried — creating drainage problems. Overhead clearances are dangerously reduced … and additional weight is added to over-passes and bridges. Cold planing is an economical method of curing all these problems.

Surface Texturing

Serious accidents increase when pavement becomes slick from wear. The textured surface produced by cold planing is highly skid-resistant and has dramatically reduced hydroplaning characteristics.

Pavement Mining

Cold milling has made it practical to actually "mine" deteriorated pavement materials from existing roads and streets. The cold planer produces an ideally-sized asphalt or concrete material which can be recycled in a variety of ways. Depending on type, age and condition of pavement, the largest cold planer can reclaim up to 900 tons of material per hour.

COLD PLANER USE BY PROJECT TYPE

Applications	Highway/Airport	City/Urban
Planing (Milling)	● To establish grade and slope. ● Remove excess pavement.	● To establish proper grade and slope. ● To establish new grade and slope.
Partial Removal	● For use with hot mix recycle. ● Remove pavement irregularities. ● Texture for skid resistance.	● To correct drainage and curb reveal. ● To lower elevation at overpass. ● For use with hot recycle. ● Eliminate leveling course.
Full Depth Removal	● Total rebuild. RAP used for base or hot recycle. ● Cold recycle. This requires additional surface treatment.	● Total rebuild. RAP used for base or hot recycle. ● Cold recycle. Requires additional surface treatment.
Texturing	● For skid resistance and improved bond when overlay is applied.	● For skid resistance and improved bond when overlay is applied.
Leveling		● At intersections to remove bumps, shoving and improve drainage.
Special	● Joint and crack repair. ● Cut rumble grooves on shoulders of bridge approaches. ● Concrete removal.	● Intersection defect repair. ● Pothole repair. ● Railroad crossing repair. ● Tight radius profiling around manhole covers, etc. ● Pavement adjustments (transitions from existing pavements to new overlays).

Rotary Mixers Overview:

The Cat Rotary Mixers combine superior performance and reliability to achieve the most demanding job specifications while maximizing machine uptime. With many enhanced features and options, the Cat Rotary Mixers are designed to work well in both full depth reclamation and soil stabilization applications.

Rotary Mixers Sustainability:

- Long-lasting lubricants and fluids extend maintenance intervals, decreasing the amount of waste and filters contributed to environment.
- Ecology drains provide a simple means to drain machine fluids with a minimized risk of spillage.
- Maintenance free Cat batteries are recyclable.
- Cat engines meet applicable emission standards.
- Rotary Mixers recycle the materials into aggregate on site. That reduces overall fuel consumption and engine emissions.
- Rotary Mixers and the reclamation process can eliminate the need for an excavator and motor grader at most job sites.
- Rotary Mixers can reclaim and stabilize simultaneously. If stabilization is required, the agent can be placed on top of the reclamation project. The Rotary Mixer then makes a single pass, mixing the stabilizer with the material beneath the surface. At the same time, it reduces the aggregate to its original size. All this is accomplished with one machine — in one pass.
- The reclamation and stabilization process can easily incorporate environmentally sensitive sealants. Now the life of the road bases can be extended without fear of harming wildlife.
- The reclamation process usually is quicker than full replacement, allowing roads to open sooner. This means fewer traffic jams — a quality of life issue as well.
- Reclamation and stabilization dramatically extend the life of roads. Some contractors conservatively estimate that a reclaimed/stabilized base will help a recycled road last 30 to 40 years. Extending that life is an enormous cost-saver, and also results in the use of less energy.

Cat RM Features:

- **Maximum Production** … from Cat engines.
- **Highly Maneuverable** … separate hydraulic pump provides hydraulic flow to large displacement motors on each rear wheel.
- **Versatility** … choice of three rotors for full depth reclamation or soil stabilization.
- **Reliability** … field proven Cat components maximize machine availability.
- **Standard all-wheel drive with advanced traction control system** for maximum traction.
- **Rotating operator's station with handwheel steering and integrated touchscreen display** provides operator with fingertip control, high visibility and exceptional comfort.

Rotor Options

Rotor	Maximum Depth of Work		No. of Bits/Tools	Direction of Cut	Stabilization	Reclamation
Universal 406 mm **(16")***	406 mm	**16"**	200	Up	X	X
Universal 457 mm **(18")****	457 mm	**18"**	200	Up	X	X
Soil	508 mm	**20"**	238	Up	X	
Combination	508 mm	**20"**	114	Up	X	
Spade	457 mm	**18"**	58	Up	X	

*Designed to produce maximum breakout force, the Universal Rotor 406 mm **(16")** performs well in severe asphalt cuts.
The Universal Rotor 457 mm **(18") is designed to provide maximum mixing depth and has lower breakout force compared to the Universal Rotor 406 mm **(16")**.

Rotor Options:

Some rotor options are not available in certain markets.

- **Soil Rotor** is designed primarily for use in soil stabilization.
- **Combination Rotor** is designed primarily for use in soil stabilization with a secondary application in light cuts of asphalt reclamation.
- **Universal Rotor** can be used for either reclamation or stabilization.
- **Spade Rotor** is designed for soil stabilization applications.

PRODUCTION ESTIMATING

The maximum cutting depth for Cat Rotary Mixers is 508 mm (20 in). In addition, the cutting width of their rotors is 2.4 m (8 ft). The following formulas allow you to determine the production in square yards (yd^2)/minute or cubic yards (yd^3)/minute.

Production in square yards (yd^2) per minute

$$yd^2/min = \frac{\text{FPM of travel speed}}{1.125}$$

$$\frac{9\ ft^2/yd^2}{8\ ft\ \text{Cutting width}} = 1.125\ \text{(This is a constant value for an eight foot wide rotor)}$$

Gallons of additive (for units with pump and metering additive system)

$$\frac{GPM}{yd^2/min} = gal/yd^2$$

Or, if required additive amounts are known, you can determine necessary travel speed as shown:

$$\frac{GPM}{gal/yd^2} = yd^2/min;\ yd^2/min \times 1.125 = ft/min$$

Production in Cubic Yards (yd^3) per minute

$$\frac{\text{FPM of travel speed}}{1.125} \times \frac{\text{Cutting or mixing depth in inches}}{36} = \frac{yd^3}{min}$$

Production in Tons per Minute

$$yd^3/min \times \frac{\text{Wt. of Material per yd in lbs}}{2000\ lb/ton} = tons/min$$

Abbreviations

FPM = Feet Per Minute
GPM = Gallons Per Minute

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WEIGHT OF MATERIALS

Material		LOOSE kg/m³	LOOSE lb/yd³	BANK kg/m³	BANK lb/yd³
Clay	— Dry	1480	**2500**	1840	**3100**
	— Wet	1660	**2800**	2080	**3500**
Clay and Gravel	— Dry	1420	**2400**	1660	**2800**
	— Wet	1540	**2600**	1840	**3100**
Sand and Gravel	— Dry	1720	**2900**	1930	**3250**
	— Wet	2020	**3400**	2220	**3750**
Sand	— Dry	1420	**2400**	1600	**2700**
	— Damp	1690	**2850**	1900	**3200**
	— Wet	1840	**3100**	2080	**3500**
Earth	— Dry Packed	1510	**2550**	1900	**3200**
	— Wet Excavated	1600	**2700**	2020	**3400**
	— Top Soil	950	**1600**	1360	**2300**
	— Loam	1250	**2100**	1540	**2600**
Bituminous Concrete	— Windrowed Chunks (25% Voids)	1740	**2925**	—	—
	— Compacted	—	—	2310	**3900**

STABILIZATION/RECLAMATION PRODUCTION

The following charts list production in square meters per minute, square yards per minute, cubic meters per minute, and cubic yards per minute. The information is based on various travel speeds and cutting depths for Cat Rotary Mixers equipped with a 2438 mm **(8 ft)** cutting rotor.

PRODUCTION RATES

Travel Speed m/min	m²/min	m³/minute Cutting Depth — mm 100	125	150	175	200	225	250	275	300	325	350	375	400	425	450	475	500
3	7.3	0.73	0.9	1.1	1.3	1.5	1.6	1.8	2.0	2.2	2.4	2.6	2.7	2.9	3.1	3.3	3.5	3.7
6	14.6	1.46	1.8	2.2	2.6	2.9	3.3	3.7	4.0	4.4	4.8	5.1	5.5	5.9	6.2	6.6	6.9	7.3
9	21.9	2.2	2.7	3.3	3.8	4.4	4.9	5.5	6.0	6.6	7.1	7.7	8.2	8.8	9.3	9.9	10.4	11.0
12	29.3	2.9	3.7	4.4	5.1	5.9	6.6	7.3	8.0	8.8	9.5	10.2	11.0	11.7	12.4	13.2	13.9	14.6
15	36.6	3.6	4.6	5.5	6.4	7.3	8.2	9.1	10.0	11.0	11.9	12.8	13.7	14.6	15.5	16.5	17.4	18.3
18	43.9	4.4	5.5	6.6	7.7	8.8	9.9	11.0	12.1	13.2	14.3	15.4	16.5	17.6	18.7	19.7	20.8	21.9
21	51.2	5.1	6.4	7.7	9.0	10.2	11.5	12.8	14.1	15.4	16.6	17.9	19.2	20.5	21.8	23.0	24.3	25.6
24	58.5	5.9	7.3	8.8	10.2	11.7	13.2	14.6	16.1	17.6	19.0	20.5	21.9	23.4	24.9	26.3	27.8	29.3
27	65.8	6.6	8.2	9.9	11.5	13.2	14.8	16.4	18.1	19.7	21.4	23.0	24.7	26.3	28.0	29.6	31.3	32.9

PRODUCTION RATES

Travel Speed ft/min	yd²/min	yd³/minute Cutting Depth — inches 4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
10	**8.9**	**0.98**	**1.2**	**1.5**	**1.7**	**2.0**	**2.2**	**2.5**	**2.7**	**3.0**	**3.2**	**3.5**	**3.7**	**4.0**	**4.2**	**4.5**	**4.7**	**4.9**
20	**17.8**	**1.96**	**2.5**	**3.0**	**3.4**	**4.0**	**4.4**	**4.9**	**5.5**	**5.9**	**6.4**	**6.9**	**7.4**	**7.9**	**8.4**	**8.9**	**9.4**	**9.9**
30	**26.7**	**2.9**	**3.7**	**4.5**	**5.2**	**5.9**	**6.7**	**7.4**	**8.2**	**8.9**	**9.6**	**10.4**	**11.1**	**11.9**	**12.6**	**13.4**	**14.0**	**14.8**
40	**35.6**	**3.9**	**4.9**	**5.9**	**6.9**	**7.9**	**8.9**	**9.9**	**10.9**	**11.9**	**12.8**	**13.9**	**14.8**	**15.8**	**16.8**	**17.8**	**18.7**	**19.8**
50	**44.5**	**4.9**	**6.2**	**7.4**	**8.6**	**9.9**	**11.1**	**12.4**	**13.6**	**14.8**	**16.0**	**17.3**	**18.5**	**19.8**	**21.0**	**22.3**	**23.4**	**24.7**
60	**53.4**	**5.9**	**7.4**	**8.9**	**10.3**	**11.9**	**13.3**	**14.8**	**16.4**	**17.8**	**19.2**	**20.8**	**22.2**	**23.7**	**25.2**	**26.7**	**28.1**	**29.7**
70	**62.3**	**6.8**	**8.6**	**10.4**	**12.0**	**13.8**	**15.6**	**17.3**	**19.1**	**20.8**	**22.4**	**24.3**	**25.9**	**27.7**	**29.5**	**31.2**	**32.8**	**34.6**
80	**71.2**	**7.8**	**9.9**	**11.9**	**13.7**	**15.8**	**17.8**	**19.8**	**21.8**	**23.7**	**25.6**	**27.7**	**29.6**	**31.6**	**33.7**	**35.6**	**37.5**	**39.6**
90	**80.1**	**8.8**	**11.1**	**13.4**	**15.5**	**17.8**	**20.0**	**22.4**	**24.5**	**26.7**	**28.8**	**31.2**	**33.3**	**35.6**	**37.9**	**40.1**	**42.1**	**44.5**

Asphalt Pavers

Sustainability:

- Long-lasting lubricants and fluids extend maintenance intervals, decreasing the amount of waste and filters contributed to environment.
- Ecology drains provide a simple means to drain machine fluids with a minimized risk of spillage.
- Maintenance free Cat batteries are recyclable.
- Cat engines meet applicable emission standards.
- Proprietary undercarriage design on Mobil-Trac™ pavers significantly reduces track belt wear, reducing frequency of replacement.
- Cat electric-heated screeds provide a reduction in engine emissions, which not only reduces the impact to the environment, but helps prevent soot build-up on indicators and decals — for enhanced safety.
- Electric-heated screeds produce less noise, which is appreciated by the crew as well as those who work and live near the construction sites.
- Electric-heated screeds, unlike their diesel counterparts, don't require a constant flame — another way emissions are reduced, and operators spared some heat.
- Electric-heated screeds are lighter, which means the paver consumes less fuel.
- Electric-heated screeds warm up quickly, another energy saver.
- Pavers equipped with Eco-mode can reduce engine speed to increase fuel efficiency during normal operation conditions.
- Pavers equipped with automatic engine speed control can be programmed to idle the engine when machine is in neutral for a predetermined amount of time, reducing fuel burn.

Features:

- **Variable width screeds** available.
- **Self-dumping hydraulic hoppers** are heavy-duty and high capacity.
- **Soldered and molded** electrical connections.
- **Self diagnostics** on propel and feeder systems.
- **Electrically heated screeds**.
- **Dual independent operating stations**.
- **Mobil-Trac undercarriage** provides speed of a wheeled paver with maneuverability of a track paver.
- **Optional Cat Grade and Slope** is integrated with the paver.
- **Large generators** deliver fast heat with less fuel consumption.*
- **One-touch functions** for cleanout/warm up mode.*
- **One-touch Auto-fill function**.*
- **One-touch feeder system activation**.*
- **Advanced generator diagnostics** helps determine heating element life.*

*Feature available on F Series pavers only.

ASPHALT PAVING CHARTS

These charts will assist you when trying to match plant output with paving speeds. Keep in mind when using these charts, it will be at 100% efficiency. If you know efficiency, multiply T.P. hour × efficiency. (Example: 75% efficiency at 300 T.P.H. – 300 × 0.75 = 225 T.P.H.)

Production in tons/hr with 1" compacted mat

Speed	Paving Widths						
ft/min	6'0"	7'0"	8'0"	9'0"	10'0"	11'0"	12'0"
10	22	26	29	33	37	40	44
20	44	51	58	66	73	80	88
30	66	77	87	99	110	120	131
40	88	102	116	131	146	161	175
50	110	129	145	164	183	201	219

Production in tons/hr with 2" compacted mat

Speed	Paving Widths						
ft/min	6'0"	7'0"	8'0"	9'0"	10'0"	11'0"	12'0"
10	44	52	58	66	74	80	88
20	88	176	116	132	146	160	176
30	132	154	174	198	220	240	262
40	176	204	232	262	292	322	350
50	220	258	290	328	366	402	438

Production in tons/hr with 3" compacted mat

Speed	Paving Widths						
ft/min	6'0"	7'0"	8'0"	9'0"	10'0"	11'0"	12'0"
10	66	78	87	99	111	120	132
20	132	153	174	198	219	240	284
30	198	231	261	297	330	360	393
40	264	306	348	393	438	483	525
50	330	387	435	492	549	603	657

Production in tons/hr with 4" compacted mat

Speed	Paving Widths						
ft/min	6'0"	7'0"	8'0"	9'0"	10'0"	11'0"	12'0"
10	88	104	116	132	148	160	176
20	176	204	232	264	292	320	352
30	264	308	348	396	440	480	524
40	352	408	464	524	584	644	700
50	440	516	580	656	732	804	876

Slope Conversion Table

Percent	Inches per foot	Inches per 12 foot	Percent	Inches per foot	Inches per 12 foot
0.17%		1/4	5.21%	5/8	7 1/2
0.35%		1/2	5.38%		7 3/4
0.52%	1/16	3/4	5.56%		8
0.70%		1	5.73%	11/16	8 1/4
0.87%		1 1/4	5.90%		8 1/2
1.04%	1/8	1 1/2	6.08%		8 3/4
1.22%		1 3/4	6.25%	3/4	9
1.39%		2	6.42%		9 1/4
1.56%	3/16	2 1/4	6.60%		9 1/2
1.74%		2 1/2	6.77%	13/16	9 3/4
1.91%		2 3/4	6.94%		10
2.08%	1/4	3	7.12%		10 1/4
2.26%		3 1/4	7.29%	7/8	10 1/2
2.43%		3 1/2	7.47%		10 3/4
2.60%	5/16	3 3/4	7.64%		11
2.78%		4	7.81%	15/16	11 1/4
2.95%		4 1/4	7.99%		11 1/2
3.13%	3/8	4 1/2	8.16%		11 3/4
3.30%		4 3/4	8.33%	1	12
3.47%		5	8.51%		12 1/4
3.65%	7/16	5 1/4	8.68%		12 1/2
3.82%		5 1/2	8.85%	1 1/16	12 3/4
3.99%		5 3/4	9.03%		13
4.17%	1/2	6	9.20%		13 1/4
4.34%		6 1/4	9.38%	1 1/8	13 1/2
4.51%		6 1/2	9.55%		13 3/4
4.69%	9/16	6 3/4	9.72%		14
4.86%		7	9.90%	1 3/16	14 1/4
5.04%		7 1/4	10.07%		14 1/2

Formula:

$$\text{Percent} = \frac{\text{Inches per foot} \times 100}{12}$$

Inches in decimals of a foot

1/16 = .0052	1 = .0833
3/32 = .0078	2 = .1667
1/8 = .0104	3 = .2500
3/16 = .0156	4 = .3333
1/4 = .0208	5 = .4167
5/16 = .0260	6 = .5000
3/8 = .0313	7 = .5833
1/2 = .0417	8 = .6667
5/8 = .0521	9 = .7500
3/4 = .0625	10 = .8333
7/8 = .0729	11 = .9167

Vibratory and Pneumatic Tire Compactors Sustainability:

- Long-lasting lubricants and fluids extend maintenance intervals, decreasing the amount of waste and filters contributed to environment.
- Ecology drains provide a simple means to drain machine fluids with a minimized risk of spillage.
- Maintenance free Cat batteries are recyclable.
- Cat engines meet applicable emission standards.
- Biodegradable oil option available.

General Compactor Features:

- **Routine maintenance** simplified by grouped service points and easy access to service areas.
- **Operator stations** designed for maximum comfort, easy control, and optimal visibility.
- **Direct hydrostatic drive to front (drums or wheels) and rear (drums or wheels)** provides dependable, responsive, propulsion effort and maximum gradeability. (Does not include pneumatic tire compactors or GC Series soil compactors.)
- **Eco-mode standard equipment** on Vibratory Soil Compactors and Double Drum Rollers reduces fuel consumption.

Vibratory Compactor Features:

Single Drum

- **Dual pump system** delivers positive tractive effort to both drum and rear wheels, regardless of underfooting. This increases the machine's ability to maneuver in a wide variety of soil types and conditions and improves gradeability. (Does not pertain to GC Series compactors.)
- **Limited slip high traction differential** is standard on all units for best traction of rear tires.
- **Optional heavy-duty front-mounted blade** with reversible cutting edge is available to allow backfilling and leveling during compaction. (Check for model availability.)
- **ROPS (Roll Over Protective Structure)** optional on all units. Enclosed cabs with EROPS rating available as an option.
- **Optional Cat Compaction Control** featuring exclusive Machine Drive Power (MDP) or CMV (accelerometer-based) measurement scalable to include GNSS mapping and recording.
- **Adjustable jaw-type cleaner bar** keeps drums clean between pads during forward and reverse movement.

Double Drum and Combi

- **Vibration automatically ceases before machine comes to a stop** to help produce a smooth, flawless mat surface.
- **Close side clearances** allow compactors to work close to curbs, walls and other obstructions.
- **Large, rust-proof water tanks and pressure spray system** provide hours of reliable operation between fill-ups.
- **Emulsion system** available for combi compactor rear tires to prevent materials sticking to tires.
- **ROPS (Roll Over Protective Structure)** available on all models. Enclosed cabs with EROPS rating available on some models.
- **Optional Cat Compaction Control** featuring GNSS mapping of pass count and mat surface temperature.

Pneumatic Tire Compactor Features:

- **All wheel oscillation.** Front and rear tires provide even wheel loads regardless of evenness underfoot.
- **High drive propel system.** Completely hydrostatic with drive motors and brakes located in mainframe away from contamination and damage.
- **Ballast compartments** are easily accessible for quick loading and are located to provide balanced wheel/weight ratio.
- **Single-lever hand control** of forward and reverse movement makes smooth rolling easy.
- **Optional Cat Compaction Control** featuring GNSS mapping of pass count and mat surface temperature.

NOTE: All models and options are not available in all markets.

Pneumatic Tires — Bias Ply and Radial

Model	Tire Size	Ply Rating	Tire Inflation Pressure			
			Minimum Pressure		Maximum Pressure	
			kPa	psi	kPa	psi
CW16	7.5 × 15	12-ply Radial	344	**50**	757	**110**
	7.5 × 15	14-ply Bias	344	**50**	862	**125**
CW34	13/80-R20	Radial	300	**44**	900	**131**

Ballast Configurations

Model	Load	Ballast Configuration					
		Empty	Water Only	Steel Only	Wet Sand Only	Steel and Water	Steel and Wet Sand
CW16	Wheel Load	580 kg **1279 lb**	970 kg **2139 lb**	940 kg **2072 lb**	1360 kg **2998 lb**	1310 kg **2888 lb**	1670 kg **3682 lb**
	Machine Weight	5200 kg **11,464 lb**	8700 kg **19,180 lb**	8500 kg **18,739 lb**	12 200 kg **26,896 lb**	11 750 kg **25,904 kg**	15 000 kg **33,069 lb**
CW16 (11-wheel)	Wheel Load	480 kg **1058 lb**	800 kg **1764 lb**	780 kg **1720 lb**	1110 kg **2447 lb**	1060 kg **2337 lb**	1350 kg **2976 lb**
	Machine Weight	5300 kg **11,685 lb**	8800 kg **19,400 lb**	8600 kg **18,960 lb**	12 300 kg **27,117 lb**	11 750 kg **25,904 kg**	14 900 kg **32,849 lb**
CW34	Wheel Load	1250 kg **2756 lb**	1620 kg **3572 lb**	2820 kg **6217 lb**	2000 kg **4409 lb**	3080 kg **6790 lb**	3375 kg **7441 lb**
	Machine Weight	10 000 kg **22,050 lb**	13 000 kg **28,660 lb**	22 550 kg **49,715 lb**	16 000 kg **35,275 lb**	24 700 kg **54,450 kg**	27 000 kg **59,525 lb**

*Configuration not available.

Ground Contact Pressures — CW16

Average Wheel Load	Tire Pressures: 12-Ply Radial						
	344 kPa 50 psi	**413 kPa 60 psi**	**482 kPa 70 psi**	**550 kPa 80 psi**	**619 kPa 90 psi**	**688 kPa 100 psi**	**757 kPa 110 psi**
545 kg **1200 lb**	236 kPa **34 psi**	266 kPa **39 psi**	284 kPa **41 psi**	306 kPa **44 psi**	317 kPa **46 psi**	317 kPa **46 psi**	344 kPa **50 psi**
970 kg **2145 lb**	250 kPa **36 psi**	284 kPa **41 psi**	314 kPa **46 psi**	343 kPa **50 psi**	369 kPa **54 psi**	378 kPa **55 psi**	410 kPa **60 psi**
1440 kg **3180 lb**	280 kPa **41 psi**	304 kPa **44 psi**	331 kPa **48 psi**	358 kPa **52 psi**	390 kPa **57 psi**	405 kPa **59 psi**	607 kPa **88 psi**

Average Wheel Load	Tire Pressures: 14-Ply Bias								
	344 kPa 50 psi	**413 kPa 60 psi**	**482 kPa 70 psi**	**550 kPa 80 psi**	**619 kPa 90 psi**	**688 kPa 100 psi**	**757 kPa 110 psi**	**826 kPa 120 psi**	**862 kPa 125 psi**
545 kg **1200 lb**	243 kPa **35 psi**	266 kPa **39 psi**	284 kPa **41 psi**	295 kPa **43 psi**	317 kPa **46 psi**	330 kPa **48 psi**	344 kPa **50 psi**	359 kPa **52 psi**	367 kPa **53 psi**
970 kg **2145 lb**	259 kPa **38 psi**	295 kPa **43 psi**	321 kPa **47 psi**	343 kPa **50 psi**	369 kPa **54 psi**	388 kPa **56 psi**	420 kPa **60 psi**	421 kPa **61 psi**	427 kPa **62 psi**
1440 kg **3180 lb**	273 kPa **40 psi**	312 kPa **45 psi**	336 kPa **49 psi**	364 kPa **53 psi**	390 kPa **57 psi**	413 kPa **60 psi**	437 kPa **64 psi**	465 kPa **68 psi**	475 kPa **69 psi**

NOTES:

1. Each tire type has a unique pressure distribution which varies with both tire inflation pressure and wheel load. The distribution of pressure along both transverse and longitudinal profiles is rarely uniform.
2. The measurements in this table represent the peak pressures measured in a transverse profile at each of the ballast conditions at maximum tire inflation pressure.
3. For most applications, it can be assumed that normal operation of the pneumatic compactor will result in the ground being subjected to pressures near the maximum during at least one machine pass.

Ground Contact Pressures — CW34

Average Wheel Load	Tire Pressure							
	300 kPa 44 psi	**400 kPa 58 psi**	**500 kPa 73 psi**	**600 kPa 87 psi**	**700 kPa 102 psi**	**800 kPa 116 psi**	**850 kPa 123 psi**	**900 kPa 131 psi**
1500 kg **3307 lb**	242 kPa **35 psi**	309 kPa **45 psi**	406 kPa **59 psi**	612 kPa **89 psi**	680 kPa **99 psi**	1038 kPa **151 psi**	1265 kPa **184 psi**	1587 kPa **230 psi**
2000 kg **4410 lb**	260 kPa **38 psi**	299 kPa **43 psi**	357 kPa **52 psi**	462 kPa **67 psi**	498 kPa **72 psi**	628 kPa **91 psi**	691 kPa **100 psi**	764 kPa **111 psi**
2500 kg **5512 lb**	308 kPa **45 psi**	322 kPa **47 psi**	360 kPa **52 psi**	429 kPa **62 psi**	458 kPa **66 psi**	539 kPa **78 psi**	577 kPa **84 psi**	618 kPa **90 psi**
3000 kg **6614 lb**	397 kPa **58 psi**	369 kPa **54 psi**	386 kPa **56 psi**	433 kPa **63 psi**	457 kPa **66 psi**	516 kPa **75 psi**	543 kPa **79 psi**	573 kPa **83 psi**
3375 kg **7441 lb**	518 kPa **75 psi**	423 kPa **61 psi**	418 kPa **61 psi**	448 kPa **65 psi**	469 kPa **68 psi**	517 kPa **75 psi**	539 kPa **78 psi**	564 kPa **82 psi**

NOTES:

1. Each tire type has a unique pressure distribution which varies with both tire inflation pressure and wheel load. The distribution of pressure along both transverse and longitudinal profiles is rarely uniform.
2. The measurements in this table represent the peak pressures measured in a transverse profile at each of the ballast conditions at maximum tire inflation pressure.
3. For most applications, it can be assumed that normal operation of the pneumatic compactor will result in the ground being subjected to pressures near the maximum during at least one machine pass.